

Comparative Study of Different MPPT Algorithms for Photovoltaic Systems under Partial Shading Conditions

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Abstract—We are interested in tracking the maximum power of a partially shaded photovoltaic (PV) system, using a DC-DC converter as the power conditioning unit. To meet this objective, many structures of power converters and different conventional and non-conventional control methods have been proposed, studied, and optimized. Among them the conventional method of perturbation and observation (P&O), and the unconventional method of particle swarm optimization (PSO), which remain the most known MPPT techniques due to their simplicity. But their dynamic response and accuracy remain low compared to other optimized methods. This work presents a comparative analysis between five control strategies: i) the conventional (P&O) method with fixed iteration step, ii) variable step (VSP&O), iii) modified adjustable step (imVSP&O), iv) The improved perturbation and global scanning and observation method (VSPO&GS), v) The unconventional method based on artificial intelligence PSO. The simulation results of these algorithms are analyzed and compared in the Matlab/Simulink environment in order to show the performances and the limits of each method in the tracking of the point of maximum power.

Keywords- photovoltaic-arrays; shading phenomenon; MPPT control; perturb and observe; particle swarm optimization.

I. INTRODUCTION

As the world's population grows, the demand for energy increases. In addition, the world's reserves of conventional energy resources will all be depleted within the next few centuries. Therefore, the interest in renewable energy sources (RES) for electricity generation is gradually increasing importance worldwide.

The solar energy source is the most attractive among all other renewable sources thanks to its durability, ease of use, and the possibility of installing it near the user for its noise-free running.

Photovoltaic cells are specific sensors that transform solar radiation into electrical energy. In order to obtain an interesting power level, it is necessary to couple several cells in series to increase the voltage, or in parallel to increase the current. However, these devices – due to the non-linearity of their output current-voltage characteristic – do not always operate at maximum efficiency under given atmospheric conditions (irradiation, temperature). The role of photovoltaic

systems is to power one or more consumers, such as smart homes, who use these systems to save energy or who are located in an area isolated from the electrical grid. A storage system associated with PV generators in smart homes, responsible for ensuring the supply at all times and for several days despite the intermittent nature of the production.

A control mechanism is required to find and track the maximum power point (MPP), called "Maximum Power Point Tracking" (MPPT). This system consists of a power converter inserted between the PV source and the load, the duty cycle of the converter is used to achieve the maximum power point tracking (MPPT). [1] To this end, several methods of monitoring the MPP have been proposed and discussed in the literature [2], such as Perturb and observe (P&O) which is the most known and debated one; It plays a crucial role in the development of photovoltaic (PV). And the biological-inspired algorithms, called non-conventional, namely the PSO method with its simplest structure and easy implementation. [3]

This paper proposes a comparative study between five algorithms: The PSO[3]–[5], the P&O[6]–[8], the VSP&O[6], [9], the VSPO&GS[10], and the imVSP&O[5]. 1. The PSO meta-heuristic algorithm can be classified in the group of optimization methods. 2. The conventional P&O algorithm, which uses a fixed iteration step (FSP&O). 3. (VSP&O) a variable step. 4. The (VSPO&GS) which is an improved method of perturbation, observation, and global scan; It tracks the maximum power point under shading that gives multi-peak characteristics of the (P-U) curve. 5. The (imVSP&O) improved variable step size MPPT algorithm that uses two different step size increments that are variable. Several comparative studies on MPP tracking are mentioned in the literature. But our analysis brings together classical and optimized methods and even artificial intelligence-based methods.

This paper is organized as follows: in section 2, the photovoltaic system modeling is presented. Section 3 is devoted to the MPPT controls design and, a comparative study between different conventional, unconventional and improved MPPT algorithms, in section 4, numerical simulation results are presented on Matlab-Simulink and then a discussion is made to evaluate the performance of the algorithms presented in standard irradiation and temperature conditions and in case of partial

shading. And finally, we summed up the main ideas of our article in the conclusion.

II. PHOTOVOLTAIC SYSTEM MODELLING

A. Study of Photovoltaic Panel

The sun is the most powerful source of energy and in addition it is free. Solar irradiation is transmitted in space uniformly and in all directions. The conversion of solar radiation into electricity is ensured by the PV effect; it was observed for the first time by Becquel in 1839.[11][12]

Photovoltaic conversion is the direct transformation of electromagnetic energy (radiation) into continuous type electrical energy that can be used directly through a semiconductor material called a photovoltaic cell.[11] The Simulink model of the solar PV and, BOOST converter and load is shown in Fig.1 .

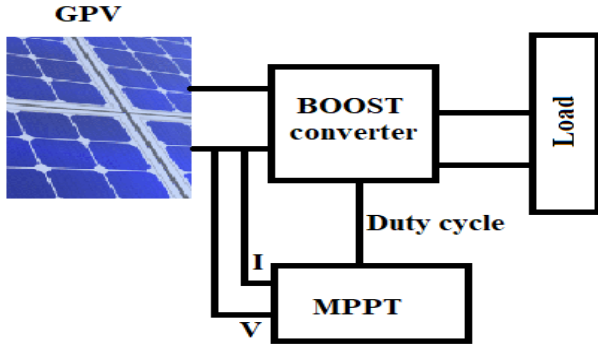


Figure 1. Model of the solar PV and, BOOST converter and load

B. Equivalent Circuits of PV Cells

The PV cell can be modeled either by using a simple diode to facilitate calculations, or using a dual diode for better accuracy, efficiency and prediction of characteristics under partial shading. [13][14]. The single-diode model, as shown in Fig.2, consists of a diode characterizing the junction, a current source characterizing the photo-current I_{ph} , a series resistor representing the joule losses R_s , and a shunt resistance R_{sh} characterizing a leakage current, R_{sh} is generally much higher than R_s .

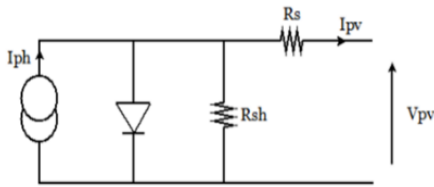


Figure 2. The single diode PV model

The relation between the current I_{pv} and the voltage V_{pv} of the corresponding circuit is in (1):

$$I_{pv} = I_{ph} - I_0 \left[\exp \left(\frac{V_{pv} + I_{pv} \cdot R_s}{\frac{n \cdot k \cdot T}{q}} \right) - 1 \right] - \frac{V_{pv} + I_{pv} \cdot R_s}{R_{sh}} \quad (1)$$

Where:

I_{ph} : Current generated by the incident light in Ampere.

I_0 : The reverse saturation current of the diode in Ampere.

K : The Boltzmann's constant ($1,38 \times 10^{-23} \text{ JK}^{-1}$)

q : The electronic load ($1,602 \times 10^{-19} \text{ C}$)

T : The cell temperature in Kelvin

R_s et R_{sh} : These respectively are the series resistance and the shunt resistance in Ω .

Fig.3 shows the equivalent circuit represented by two exponential diodes. One is an ideal diode, while the other is not ideal.

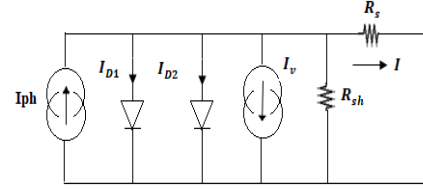


Figure 3. The dual diode PV model

We apply Kirchhoff's law to obtain the relation between the current and the voltage of the PV system:

$$I = I_{ph} - I_{D1} - I_{D2} - I_v - I_{sh} \quad (2)$$

$$I_{D1} = I_{s1} \left\{ \exp \left[\frac{q(V + R_s I)}{A_1 K T} \right] - 1 \right\} \quad (3)$$

$$I_{D2} = I_{s2} \left\{ \exp \left[\frac{q(V + R_s I)}{A_2 K T} \right] - 1 \right\} \quad (4)$$

$$I_{sh} = \frac{V + R_s I}{R_{sh}} \quad (5)$$

Where:

I_{D1} et I_{D2} : Diode current

I_{s1} et I_{s2} : Saturation current of the diode.

The current-voltage (IV) and power-voltage (PV) characteristics of the solar PV model adapted in this paper under standard test conditions are shown in Fig. 4. Concerning the (IV) and (PV) characteristics under partial shading conditions, they are shown in Fig.5, depending on environmental conditions (irradiation, $I_{rr} = 1000 \text{ W/m}^2$ and temperature, $T = 25^\circ \text{C}$). The set of five values that characterize the equivalent circuit of PV cells are reported in Table.1.

TABLE I. PARAMETERS OF THE PV ARRAY

Name	Parameters	Values
Light-generated	I_{ph}	8.6307 A
Diode saturation	I_0	$1.4176e^{-10}$ A
Diode ideality	n	0.99132
Series resistance	R_s	0.098625 Ω
Shunt resistance	R_{sh}	82.1161 Ω

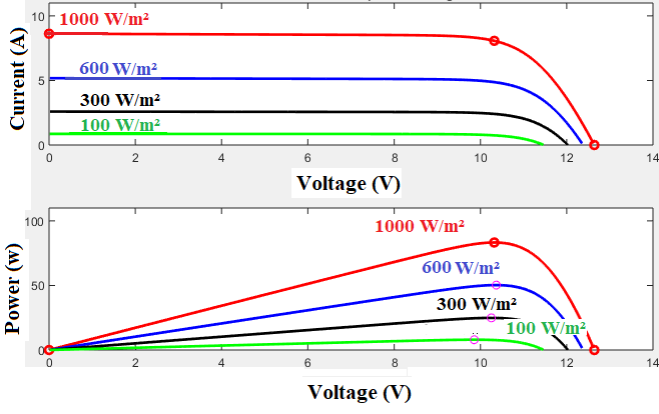


Figure 4. I-V and P-V characteristics of a PV model under standard test conditions

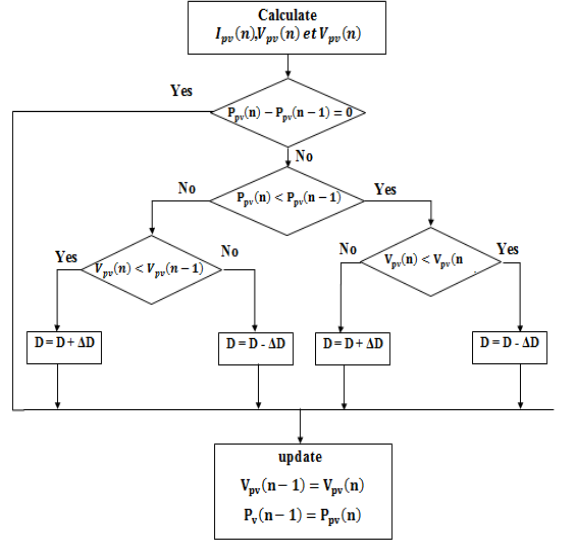


Figure 6. P&O algorithm flowchart

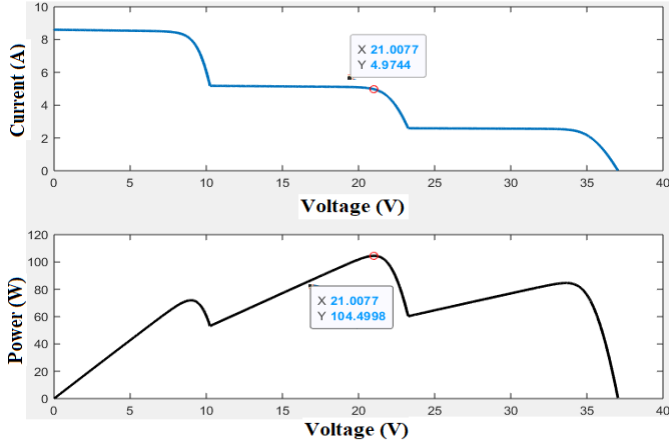


Figure 5. I-V and P-V characteristics of the partially shaded PV model

III. THE MPPT TECHNIQUES

The MPPT control system is an essential link in photovoltaic power generation systems; therefore, many conventional and non-conventional MPPT algorithms have been applied on different occasions, and under different environmental conditions.

A. Perturb and Observe (P&O) Algorithm

The mechanism of the P&O algorithm in Fig. 6 works by disturbing the voltage or current across the PV array, then comparing the corresponding output power with that of the previous cycle, in order to define the variation (increment or decrement) of the solar array voltage or current, or the variation of the duty cycle (D) until the MPP is reached. The disadvantage of this technique is that the output power starts to fluctuate on the MPP with a fixed step until the end of the procedure. It should be noted that selecting a larger step size can result in a fast response, but at the cost of higher power losses. Conversely, selecting smaller values for the step size results in a slower response, but with reduced power losses.[7]

B. Variable Step P&O Algorithm

As shown in Fig.7, the principle of this algorithm is that when the operating point is far from the MPP, the algorithm increases the increment size. While, when the operating point is close to the MPP, the step size increment becomes too small; therefore, the oscillation becomes very small around the MPP, which helps to increase the efficiency of the PV system [5]. In [15] the adopted variable step size is presented in (6) :

$$\text{step} = N \left| \frac{dP}{dV} \right| \quad (6)$$

N is the scaling coefficient defined in the design to control the step size. Therefore, the size of the step must satisfy:

$$N < \Delta D_{\max} < \left| \frac{dP}{dV} \right| \quad (7)$$

$\Delta D_{\max}$: The largest disruption of the duty cycle allowed by the P&O algorithm.

The range of the scale factor N is determined in the previous equation. The step size takes the maximum value of the fixed step $\Delta D_{\max}$ defined earlier. When the solar irradiation is rapidly changed, similarly (7) shows a comparison of the power and voltage variation $\frac{dP}{dV}$ at a constant ($\Delta D_{\max}$ and N). Then, under fast and high irradiation changes, it is not possible to find an appropriate scaling coefficient (N) and a large step limiter ($\Delta D_{\max}$) that meets the requirements of the MPPT system.[16]

A. Improved P&O Algorithm with Variable Step Size (imVSP&O)

In order to overcome the problem mentioned in the last paragraph regarding N and $\Delta D_{\max}$, and if the line ($\Delta D_{\max} / N$) moves up and down when the radiation level of the sun changes, the above mentioned problems will disappear.[16]

This new algorithm, in Fig. 8, uses two different step size increments and these steps are variable. This technique measures

the current variation (ΔI) to determine which of the two steps (step 1 or step 2) defined in (8) and (9) is used to track the MPP.

Equation (6) is multiplied by a normalization coefficient (A_1), in case of an increase in the solar irradiation level that increases the current variation ΔI , the step size coefficient N' increases. But, if this irradiation level decreases, the current decreases, resulting in a reduction in the step size, and therefore, the equation (6) is multiplied by a new normalization coefficient A_2 .[5]

$$\begin{cases} \text{Step 1} = \left(\frac{N'}{I} \left| \frac{dP}{dV} \right| \right) A_1 \\ \text{Step 2} = \left(\frac{N'}{I} \left| \frac{dP}{dV} \right| \right) A_2 \end{cases} \quad (8)$$

$$(9)$$

With: $A_1 < 1 < A_2$

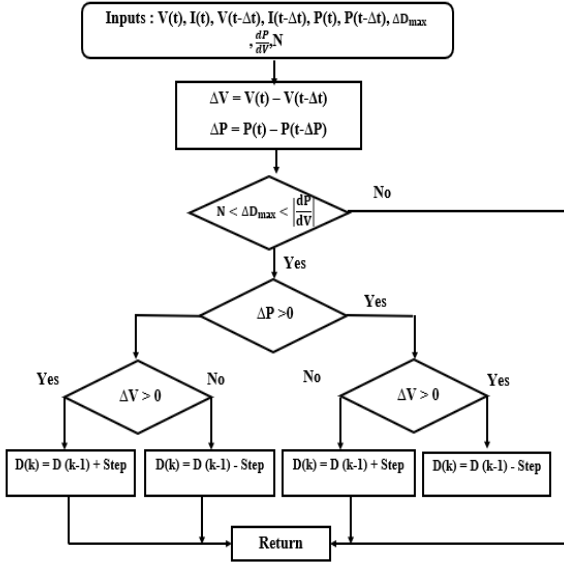


Figure 7. Flowchart of the variable step P&O algorithm (VSP&O)

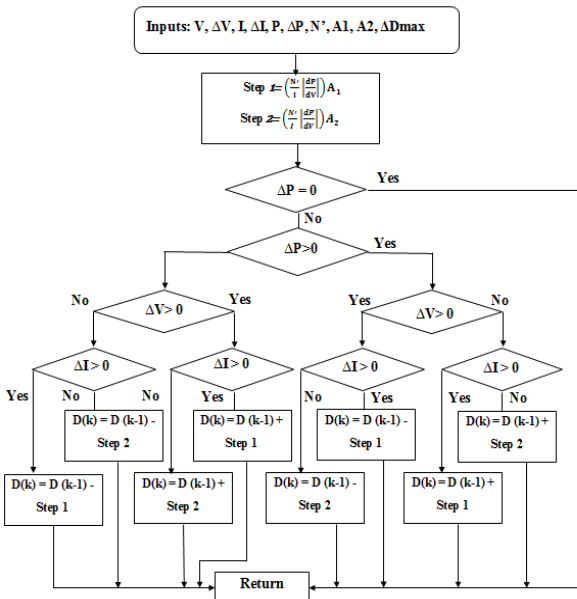


Figure 8. Flowchart of the improved VS_PO algorithm (imVSP&O)

C. Improved Variable Step P&O with Global Scanning (VSPO&GS) Algorithm

This method adopts the duty cycle to operate in two steps: The first is based on the variable step P&O algorithm presented in Fig.7 to get the first peak U_1 , and the power P_1 will be recorded as P_{max} . The algorithm follows the search to find the second peak U_1 and P_2 , and then it makes the comparison If $P_2 > P_{max}$, then $P_{max} = P_2$; otherwise P_{max} remains the same P_1 . Until the last peak, P_n and P_{max} are updated at the point of maximum power. And the second step of global scanning is used to get the global optimum. The (VSPO&GS) algorithm Fig.9 guarantees that it will not fall into a local optimum and the power oscillation will be reduced.[10]

The restart condition of the algorithm: When there are changes in the irradiance, the output power of the photovoltaic panel changes. Therefore, it is necessary to restart the VSPO&GS algorithm to make the PV system to find the MPP and work in a stable way. The simulation results show that the power change is higher than 15W when the shading conditions change. So the restart conditions are defined as follows [10]:

$$|P_r - P| > 15 \quad (10)$$

Where: P_r is the output power of the photovoltaic panel operating in real time, P is the recorded power.

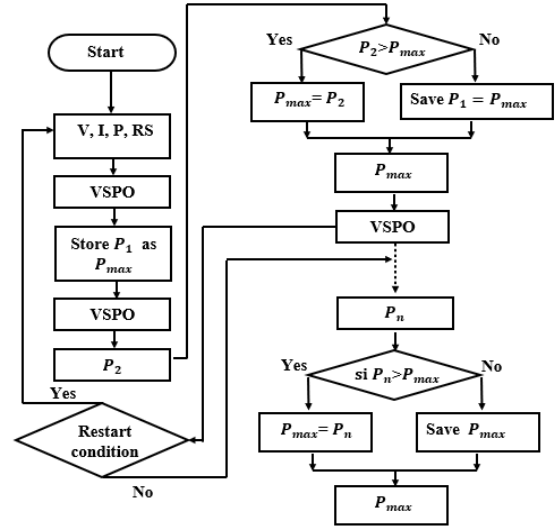


Figure 9. Flowchart of the VSPO&GS algorithm

D. Particle Swarm Optimization (PSO) Algorithm

Particle swarm optimization (PSO) is an unconventional, simple and efficient meta-heuristic approach. It was developed by Eberhart and Kennedy in 1995[17], this method solves the problems of complex and non-linear systems with a set of particles [18], The movement of these particles in the is illustrated in Fig. 10 [19].

The position of each particle is defined by (11):

$$x_i^{k+1} = x_i^k + a_i^{k+1} \quad (11)$$

The PSO technique considers a number of particles characterized by their position x_i and their velocity a_i , in a research space. At first, the particles are perturbed indiscriminately or randomly in the search space. At each

iteration (k+1), the particles move and use two final values: P_{best} which represents the best personal position and G_{best} which represents the best position of the group, in order to update their positions x_i and their accelerations a_i [20], [21] according to (12) given below:

$$a_i^{k+1} = w a_i^k + c_1 r_1^k (P_{best_i}^k - x_i^k) + c_2 r_2^k (G_{best_i}^k - x_i^k) \quad (12)$$

Where a_i^k is the acceleration of particle i after k times of iterations, w is the inertia weight, c_1 and c_2 are the acceleration coefficients for moving to each individual or global between $[0; 2.05]$, respectively r_1 and r_2 are random numbers between 0 and 1. [20] In our study and to find the MPP, the position of the particles and their fitness are considered as the duty cycle and output power of the PV system respectively. The position x_i of the particle is updated by the previously mentioned relations (11) and (12). [22] Fig.11, shows the flowchart of the PSO. [23]

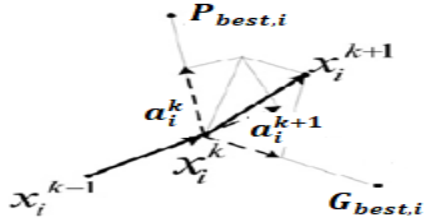


Figure 10. Movement of particles in a search space

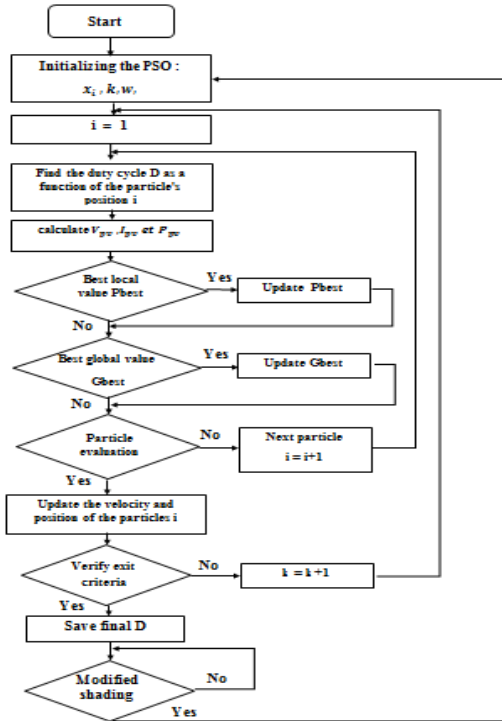


Figure 11. Flowchart of the PSO algorithm

IV. SIMULATION RESULTS AND DISCUSSION

The partially shaded PV solar system has been tested with the five maximum power point algorithms motioned previously. The total 60 PV cells are divided into three models and each model has 20 cells. The 20 cell model is evaluated with the following parameters: The open circuit voltage is 12.64 volts, voltage at maximum power is 10.32 volts, short circuit current is 8.62 A, the current at maximum power point is 8.07 A, and the maximum power point is 83.2824 watts. These three models are connected in series and each model has a bypass diode. The Simulink model of the three solar PV systems connected in series is shown in Fig.12.

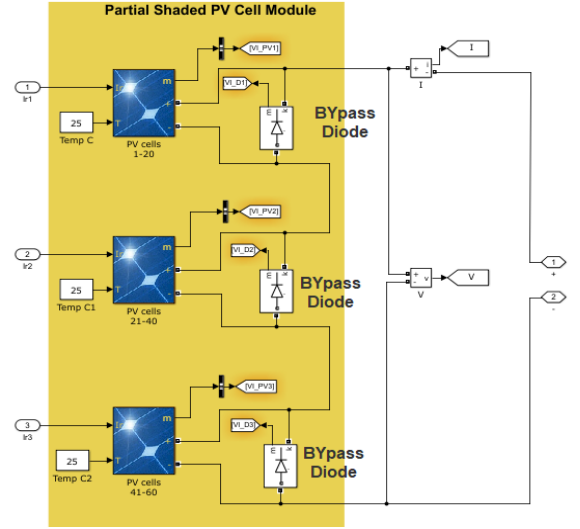


Figure 12. Model of three PVs connected in series

The partial shading condition of the solar PV model is created by setting the model irradiance (Irr) to different values, Irr1: 1000 W/m², Irr2: 600 W/m², Irr3: 300 W/m², Irr4: 100 W/m², the T = 25°C, the load resistance, R = 30 Ω, and the maximum power point is 104 Watts, this point is extracted by a boost converter connected between the solar photovoltaic panel model and the load.

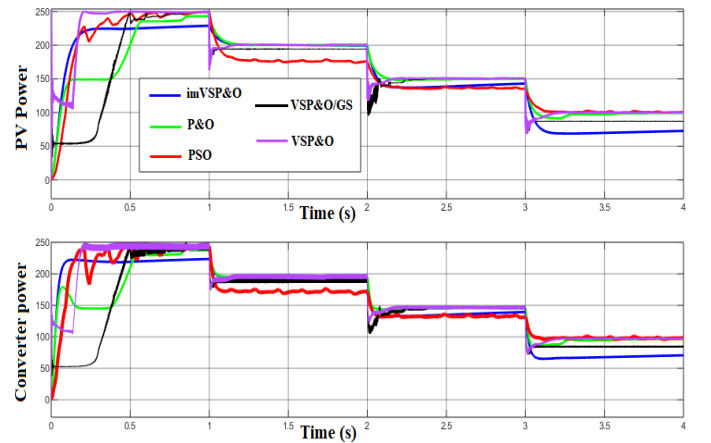


Figure 13. PV power and converter power under partial shading conditions

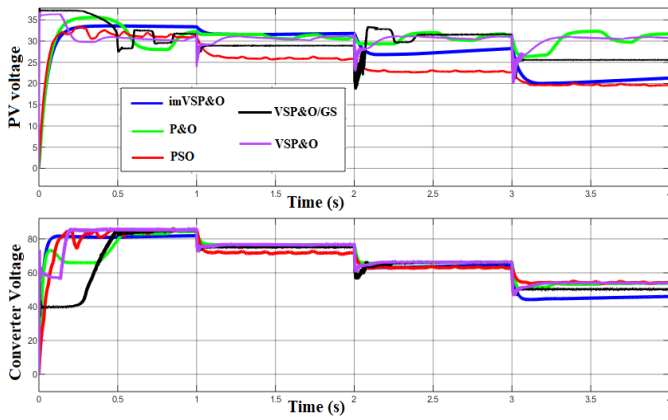


Figure 14. PV voltage and converter voltage under partial shading conditions

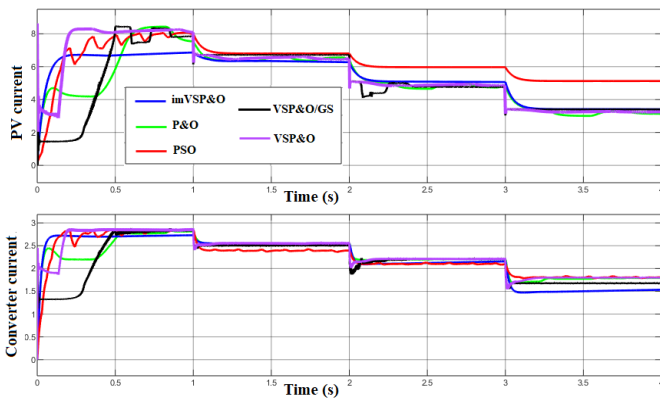


Figure 15. PV current and converter current under partial shading conditions

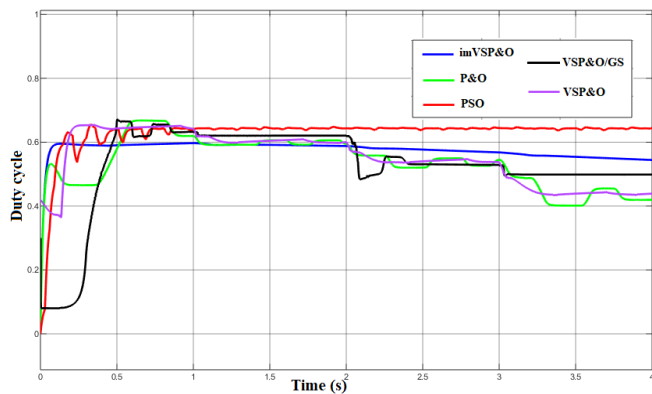


Figure 16. Duty cycle at different irradiation and constant temperature

Fig.13, 14, 15 and 16 represent respectively the power, voltage, current and the duty cycle of the PV panel and the converter. In the first-time interval the solar irradiation is set in 1000 W/m^2 , the maximum power produced by the generator (PV) $P = 250 \text{ W}$ according to the analysis, the P&O, VSP&O/GS and PSO take more time to reach the maximum power point, (400 to 600 ms). But the imVSP&O extract the maximum power of the photovoltaic panel quickly (less than 200 ms), the VSP&O method remains in maximum with a good response time. For the second interval, when the solar irradiation has been increased to 600 W/m^2 , the generator provides a maximum power of

$P = 200 \text{ W}$ given by the VSP&O, P&O and imVSP&O methods in an acceptable response time, if the irradiation is 300 W/m^2 , the maximum power found is $P = 150 \text{ W}$ by the P&O algorithm in a faster response time than the VSP&O method, the P&O and VSP&O/GS in an acceptable response time. For the last interval after a decrease of the solar radiation to 100 W/m^2 the power decreased to $P = 100 \text{ W}$ given by the VSP&O and PSO methods with an important response time compared to the other methods, while the imVSP&O and her maximum power did not exceed $P = 70 \text{ W}$. Therefore, the algorithms based on artificial intelligence are characterized by their robustness and efficiency. Indeed, they work at the optimal point without oscillations, and especially in transient state. Except that the implementation of this type of algorithm is more complex than the classical algorithms. However, the algorithms based on artificial intelligence (PSO) are characterized by their robustness and efficiency. Indeed, they work at the optimal point without oscillations, and especially in transient state. Except that the implementation of this type of algorithm is more complex than the classical algorithms.

V. CONCLUSION

In this report, we have presented the main elements of the PV system. Then, we conducted a comparative study between different conventional, unconventional and improved MPPT controls. Finally, we ended with a simulation of these algorithms. The results of the simulations show that the algorithm (VSP&O) gives more important results than the others. This algorithm is based on the variation of the step size, which reduces the oscillations around the MPP, with a faster response to reach it, when a change in solar irradiation occurs, the step size is automatically adjusted according to the duty cycle.

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